Descriptive Statistics and Technical Interpretation

Software / Machine Learning Domain

Submitted By

Snigdha Routray

Registration Number: 240962442

Repository

GitHub Repository:

https://github.com/SnigdhaRoutray/Synergy_TP

Submission Date

03 July 2026

Manipal Institute of Technology

1. Descriptive Statistics and Why They Matter

Descriptive statistics are numerical values which summarize a large number of measurements to a smaller number of meaningful values such as average, dispersion, extremes, or distribution of the data. They only describe the data that is already there.

These steps are performed before any form of cleaning, visualization, feature engineering, or modelling since it is impossible to make appropriate choices regarding a dataset which you have not analysed yet.

Statistics tell us what the data looks like, which is the mean, range, minimum, maximum, standard deviation. It does not give information about the reasons for the result obtained. For instance, the average voltage in a circuit, which is calculated to be 11 V, is only an observation. No information can be inferred from that value regarding stability of the system due to its design, instability due to defective equipment, or dependence on room temperature.

Consider a data series of 100 pH measurements from a chemical solution that is taken during the same period of 10 minutes with the help of the same probe. The average pH is 5, and the standard deviation is small enough for the figures to seem uniform. However, as each measurement was made during the same period of time by the same device, the small variance does not provide us with a reliable estimation of the stability of the solution. We cannot confirm that the solution is constant just because we have not detected any variation in its pH.

2. Center and Spread

Mean refers to the sum of numbers divided by the number count. **Median** is the middle value obtained when the data is arranged. **Minimum** and **maximum** refer to the lowest and highest values respectively, while **Range** refers to the difference between the minimum and maximum values. **Variance** is used to measure the average squared distance from the mean, while **Standard Deviation** is the square root of variance. The **IQR** (interquartile range) represents the spread of the middle 50% of the data, which is the difference between the 75th and 25th percentiles.

Center (mean, median) tells you where the data points lie. **Spread** (variance, standard deviation, range, IQR) tells you how consistent or scattered the data points are. Two datasets can have the same mean but be totally different in terms of their spread.

The mean becomes unreliable where there are outliers or a skewed distribution in the data set. If you have just one outlier value, for example a sensor that outputs 500V when other values are 12V, the mean would be distorted despite the fact that this value does not show normal activity of the sensor. Median is better to use in such cases as it not influenced by the size of the data but influenced by its position only.

A large value for standard deviation in experimental readings normally indicates an inconsistent process, a noisy instrument, or variable conditions from one run to another. The IQR statistic is particularly valuable for a skewed distribution or an outlier, as it considers only the central 50%, not the outer 25% at each end.

3. Distribution Shape and Sample Size

A **distribution** is a pattern which gives us information about the distribution of values over their entire range. It tells us whether data is clustered around one point, whether it is evenly distributed, or whether they are distributed in only one tail.

The measure of **skewness** helps understand the asymmetry of the distribution of the set of data. This shows whether data is balanced or it is concentrated on one side. **Cluster** refers to a bunch of data points that are clustered together and are separated from other bunches. **Gap** refers to a wide interval of data points in which there is no data present at all. **Spread** (or dispersion) tells us how spread out your data is.

A statistic calculated using five observations is unreliable since small sample sizes tend to be highly variable. An odd observation could make the mean or variance extremely different, and we cannot know whether it is meaningful or coincidental. A statistic calculated using five hundred observations smooths out randomness and represents the underlying process better. The sample size needs to be considered when dealing with any means, variances, or patterns since small sample sizes may lead to statistics that appear meaningful yet are nothing but noise.

Example: Suppose you test the tensile strength of an alloy and find that three tests conducted on the alloy show unusually high load at which they broke. You can assume from this that the alloy has unusually high tensile strength. If you tested the same tensile strength of the alloy using fifty samples, then you would see that three samples were exceptional, while the others were average.

4. Outliers and Domain Context

An **outlier** is an observation which lies far away from the other values of the data set.

Outliers do not always signify errors. They may point to rare events, special cases in sensing or just unusual but correct conditions. If there is no concrete proof that an outlier was due to a mistake in any of the above ways, then it must be investigated rather than immediately removed because doing so might mean losing valuable data.

- In Biochemistry, a pH value of 2.1 for a solution where neutrality should be observed indicates either contamination, using an incorrect buffer, calibration error, or even acidification process.
- In Electronics, a sudden change in current within an electric circuit, which operates with stable load, can be caused by inrush currents, short circuit or even a real load surge from the connected component.
- In Mechanics, unexpectedly high value of force measured at loading may be caused by real higher strength of material, units difference (N vs kN), saturated sensors or even an unusual structure phenomenon.

5. Plot Selection and Interpretation

- **Histogram:** Used to see how the values of one set of numbers are spread out.
  Example: Measure the heights of 100 students and see how many students fall into different height ranges.
- **Box Plot:** Used to compare different groups and quickly see which group has a higher middle value, more spread, or unusual values (outliers).
  Example: Compare the test scores of Class A, Class B, and Class C.
- **Bar Chart:** Used to compare the values of different categories.
  Example: Compare the number of students in different school clubs (Sports, Music, Art, Science).
- **Scatter Plot:** Used to see if two numerical values are related.
  Example: Compare the number of hours students' study with their exam scores to see if studying more leads to higher marks.
- **Line Plot:** Used to show how something changes over time or in order.
  Example: Record the temperature every hour during the day to see how it changes.

Every graph or plot should have a clear title, axis labels, and units because they help people understand the data correctly. The **Title** tells the reader what the graph is about. **Axis Labels** show what each axis represents. **Units** are used to convey the measurement being used. Units make the graph easy to understand and prevent confusion or misunderstanding. Table 1. shows the usage of different charts in different domains.

Chart Type	**Biochemistry Example**	**Electronics Example**	**Mechanical Example**
Histogram	Distribution of enzyme activity	Distribution of resistor values	Distribution of bolt diameters
Box Plot	Compare pH values of different samples	Compare voltage of different circuits	Compare strength of different materials
Bar Chart	Compare protein concentrations in samples	Compare power consumption of devices	Compare tensile strength of different alloys
Scatter Plot	Enzyme concentration vs. reaction rate	Voltage vs. current	Force vs. deformation
Line Plot	Cell growth over time	Current over time	Temperature of an engine over time

Table 1. Usage of various charts in different fields

6. Correlation, Causation, and Confounding

Correlation refers to two variables moving together in some way - where an increase or decrease of one variable brings about the same change in another variable. **Causation** refers to the effect of change in one variable bringing about the effect of change in another variable. Correlation cannot be taken as causation since both variables may be affected by something completely different.

The third variable is referred to as the **confounding variable** which affects both variables being tested and creates correlation among the two but not direct relationship between the two variables.

In student related data, a relationship may exist between number of hours slept and marks scored in exams, but this does not mean that sleep causes good performance since a confounding factor such as discipline while studying could be causing the two to correlate. The same applies to mechanical data where ambient temperature and failure of machine parts may be correlated but the real cause could be some lubricant that thins due to temperature increase.

7. Domain Measurement Interpretation and Comparability

Measurements in pH, voltage, current, force, and pressure cannot be averaged since these measurements refer to different properties with different units. Therefore, the mean would not make sense. The units are critical in a measurement because one value can refer to different things. For instance, 10 V is different from 10 N. The values need to be grouped by the domain, measurement type, unit, and experimental condition before data interpretation such that only comparable data are considered. Data that can be compared include voltage in the same circuit for different loads, force in the same material at different temperatures, and pH in the fermentation batches under the same conditions. Data that cannot be compared include pH versus voltage, the average of force that was recorded in newtons and pounds without converting the units, and the comparison of currents from different experimental conditions. The following information should be sought from domain teams prior to data analysis: the units of measurement, experimental condition, sample size, and whether there were any errors during the measurement process.

8. Final Reflection

Even if statistics is calculated correctly, it does not necessarily provide the right message if the context is missing for the data. The value can only become meaningful when we know what this value measures and in which units. Thus, for instance, the value of 5 may measure 5V or 5 Nand it is impossible to compare these values since they refer to different things. Similarly, the average of pH, voltage, and force will not make sense scientifically. Moreover, a small sample size cannot represent the whole picture of the system and thus any conclusion made from such data cannot be accurate. Outliers need to be explored instead of being simply discarded as they may reveal some important events like malfunctioning of instruments or contamination. The correlation of two variables does not mean that there is a cause-effect relationship since other factors can influence both these variables. It is also crucial to know about experimental conditions, calibration of instruments, measurement techniques, and units used before making an analysis.

Review Questions

1. **If mean and median are far apart, what does that suggest?**
   The data may be skewed or have outliers.
2. **Why isn't standard deviation enough without unit and context?**
   The value has no meaning without knowing what is being measured and its unit.
3. **Is every outlier wrong?**
   No. Some outliers may be real and important, not errors.
4. **Why is 105% attendance a valid data type but an invalid value?**
   It is a number (percentage), but attendance cannot be more than 100%.
5. **Why is an average of pH, voltage, current, and force meaningless?**
   They measure different things with different units.

6. **Which is more robust to outliers: mean or median?**
   Median, because it is less affected by extreme values.
7. **What does a box plot show that a bar chart doesn't?**
   A box plot shows the spread, median, and outliers in the data.
8. **If attendance and score have positive correlation, what can and can't you conclude?**
   They increase together, but you cannot say attendance causes higher scores.
9. **Give an example of a statistically correct but scientifically useless statement.**
   "The average of pH, voltage, and force is 14."
10. **What extra metadata should be requested before analysis?**

- Units of measurement
- Experimental conditions
- Sample size
- Instrument calibration
- Any known errors or problems in data collection

References

[1] D. Diez, C. Barr, and M. Çetinkaya-Rundel, *OpenIntro Statistics*, 4th ed. OpenIntro, 2019. [Online]. Available: https://www.openintro.org/book/os/ (Biostatistics)

[2] Khan Academy, "Statistics and Probability." [Online]. Available: https://www.khanacademy.org/math/statistics-probability (Khan Academy)

[3] National Institute of Standards and Technology (NIST), *NIST/SEMATECH e-Handbook of Statistical Methods*, "Exploratory Data Analysis." [Online]. Available: https://www.itl.nist.gov/div898/handbook/ (Investopedia)

[4] StatQuest, "StatQuest with Josh Starmer." [Online]. Available: https://statquest.org/

[5] An Introduction to Statistical Learning, G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, *An Introduction to Statistical Learning: with Applications in Python*, 2nd ed. Cham, Switzerland: Springer, 2023.